



OFFLINE-FIRST SECURITY ENGINEERING

VNAGY — Modular Offline-First Security Doctrine

R&D Concept Paper (Full Version)

Document ID: VNAGY-FSD-001

Version: v1.0

Revision History:

Version	Date	Author	Description
V1.0	1 September 2026	V. Nađ	Draft

Document Status: Internal Technical Specification

Prepared by: VNAGY Security Engineering

Reviewed by: V. Nađ — Internal Peer Review

Approved by: V. Nađ — Standards Committee

Location: Osijek

Contact: viktorijanad8@gmail.com

Licensed under VNAGY CC BY-NC 4.0 Extended License. Short documents may use VNAGY Minimal License (PSDL-1.3).

© Viktorija Nađ, 2026 — All Rights Reserve

Table of Contents

1. Abstract.....	1
2. Problem Statement.....	2
3. Core Doctrine Principles.....	3
3.1 Offline-First.....	3
3.2 Passive Isolation.....	3
3.3 Modular Boundaries.....	3
3.4 Behavior-First Detection.....	3
3.5 Governance Shield.....	3
4. VNAGY Architecture Overview.....	4
4.1 3D Modular Network.....	4
4.2 Behavior Engine.....	4
4.3 Heuristics & Scoring Model.....	5
4.4 Killing Chain Integration.....	5
4.5 Governance Shield.....	5
5. Why VNAGY Works (Doctrine Layer).....	6
5.1 Isolation Works Because AI Attacks Depend on Visibility.....	6
5.2 Offline-First Works Because Cloud Coordination Amplifies Attacks....	6
5.3 Behavior Detection Works Because Behavior Cannot Be Faked Consistently.....	6
5.4 Modular Boundaries Work Because They Remove Escalation Paths...6	
6. Application Domains.....	7
6.1 Industrial OT.....	7
6.2 IoT & Edge.....	7
6.3 Embedded & Firmware.....	7
6.4 SIEM Export.....	7
6.5 CPU Behavior Monitoring.....	7
7. VNAGY Rule Format.....	8
8. Governance Shield (Detailed).....	9
9. Conclusion.....	10
10. Intellectual Property & Licensing.....	11
10.1. Ownership.....	11
10.2. Permitted Use.....	11
10.3. Prohibited Use.....	11
10.4. Non-Operational Declaration.....	11
10.5. License.....	12

1. Abstract

Modern systems expose too much surface area. AI-driven attacks exploit full-system visibility, cloud feedback loops, and memory-only behavior to escalate rapidly. VNAGY introduces a **modular, offline-first security doctrine** designed to eliminate systemic exposure by isolating sensitive components inside strict boundaries. The doctrine replaces reactive defense with structural protection, using behavior-based detection, passive isolation, and governance-driven integrity.

VNAGY

2. Problem Statement

Contemporary architectures rely heavily on cloud telemetry, centralized coordination, and shared-state components. This creates three critical weaknesses:

- **Full-system visibility** enables AI attacks to map the entire environment.
- **Cloud feedback loops** amplify escalation paths.
- **Memory-only attacks** bypass signature-based detection.

Traditional defense models assume that systems must remain connected, observable, and reactive. **VNAGY** challenges this assumption.

VNAGY

3. Core Doctrine Principles

3.1 Offline-First

Systems operate locally, without cloud dependency. Offline-first removes coordinated escalation, reduces exposure, and ensures that detection logic cannot be manipulated externally.

3.2 Passive Isolation

Modules do not share memory, runtime, or state. Isolation is structural — not a configuration. Passive isolation eliminates the attack surface instead of defending it.

3.3 Modular Boundaries

Each module sees only what it must see. Boundaries break lateral movement and prevent privilege escalation.

3.4 Behavior-First Detection

Detection is based on CPU behavior, heuristics, scoring, and killing-chain signals — not signatures or cloud telemetry.

3.5 Governance Shield

A local integrity layer that enforces rules, boundaries, and policy. Governance Shield ensures that doctrine principles are consistently applied across all modules.

4. VNAGY Architecture Overview

- Components **MUST NOT** access external networks.
- Modules **MUST NOT** share internal operational state.
- AI-adjacent logic **MUST NOT** initiate cross-module communication.
- No component **MUST NOT** expose thresholds, algorithms, or dynamic logic.

4.1 3D Modular Network

VNAGY structures the system as a multi-layered network of isolated modules. Sensitive components operate inside isolation barriers, protected from probing and escalation.

Modules include:

- **Behavior Engine**
- **Governance Shield**
- **Network Security Layer**
- **Embedded/Firmware Layer**
- **IoT & Edge Layer**
- **SIEM Export Layer**

Each module is independent, offline-first, and governed by strict boundaries.

4.2 Behavior Engine

The Behavior Engine evaluates:

- CPU behavior
- memory-only activity
- heuristics
- scoring
- killing-chain signals

It does not rely on cloud telemetry. It interprets behavior locally, using deterministic heuristics.

4.3 Heuristics & Scoring Model

Heuristics define how signals are interpreted. Scoring determines whether behavior crosses thresholds. This creates a dynamic, adaptive detection model that cannot be bypassed by signature evasion.

4.4 Killing Chain Integration

VNAGY uses a custom killing chain aligned with MITRE phases. Each phase is mapped to behavior signals, enabling early detection of escalation attempts.

4.5 Governance Shield

Governance Shield enforces:

- module boundaries
- integrity checks
- rule format
- policy
- compliance
- local pipeline
- storage integrity

It is the doctrinal layer that ensures consistency across the entire system.

5. Why VNAGY Works (Doctrine Layer)

- Symbolic scoring **MAY** be performed locally.
- Deterministic transitions **MAY** be logged.
- Static configuration **MAY** be referenced.

5.1 Isolation Works Because AI Attacks Depend on Visibility

AI attacks require full-system visibility to coordinate escalation. Isolation removes visibility, breaking the attack's cognitive model.

5.2 Offline-First Works Because Cloud Coordination Amplifies Attacks

Cloud telemetry creates a feedback loop that attackers exploit. Offline-first eliminates this loop entirely.

5.3 Behavior Detection Works Because Behavior Cannot Be Faked Consistently

Signatures can be bypassed. Behavior cannot.

5.4 Modular Boundaries Work Because They Remove Escalation Paths

Lateral movement becomes structurally impossible. Privilege escalation becomes improbable.

6. Application Domains

6.1 Industrial OT

Passive isolation protects critical infrastructure from coordinated attacks.

6.2 IoT & Edge

Offline-first modules prevent cloud-driven compromise.

6.3 Embedded & Firmware

Low-level behavior monitoring detects silent processes and memory-only attacks.

6.4 SIEM Export

Local logs and offline pipelines ensure integrity and prevent tampering.

6.5 CPU Behavior Monitoring

Memory-only detection becomes native, not an add-on.

7. VNAGY Rule Format

The **VNAGY** Rule Format defines:

- behavior signals
- scoring thresholds
- module boundaries
- integrity rules
- killing-chain mapping

It is the doctrinal standard that governs how **VNAGY** interprets behavior.

VNAGY

8. Governance Shield (Detailed)

Governance Shield ensures:

- integrity
- policy enforcement
- boundary protection
- rule consistency
- local pipeline control
- storage verification

It is the doctrinal layer that protects the doctrine itself.

VNAGY

9. Conclusion

VNAGY introduces a modular, offline-first doctrine for modern security systems. It replaces cloud dependency with local intelligence, replaces exposure with isolation, and replaces reactive defense with structural protection. **VNAGY** is not an operating system — it is a **security doctrine** that defines how a system thinks, reacts, and isolates itself.

VNAGY

10. Intellectual Property & Licensing

10.1. Ownership

The **VNAGY — Modular Offline-First Security Doctrine**, including all symbolic constructs, interaction rules, structural definitions, and conceptual models, is the exclusive intellectual property of the author.

All rights are reserved.

10.2. Permitted Use

The following actions are permitted:

- reading and personal study
- academic reference
- citation with attribution
- sharing the document in its original, unmodified form

10.3. Prohibited Use

The following actions are strictly prohibited:

- implementation of any construct, rule, or model
- derivation of operational logic from symbolic definitions
- reconstruction of algorithms, thresholds, weights, or detection semantics
- commercial use in any form
- modification, removal, or alteration of licensing terms
- redistribution of modified versions

10.4. Non-Operational Declaration

This document contains no operational details, no thresholds, no weights, no heuristics, no correlation logic, and no algorithmic semantics. All constructs are symbolic and conceptual. The model is non-computational and non-reconstructable by design.

10.5. License

VNAGY Extended License (PSDL-1.3)

VNAGY CC BY-NC 4.0 Extended License

© 2026 Viktorija Nađ — All rights reserved.

License reference:

<https://github.com/VNAGY-sec-ViktorijaN/VNAGY-Framework/tree/main/VNAGY/LICENSE>

VNAGY